## Divisibility by 9

**A number is divisible by 9 if the sum of its digits is divisible by 9.** Note that the numbers divisible by 9 are also divisible by 3.

**Examples:** 9567 is divisible by 9 as 9+5+6+7=27 is divisible by 9.

## Divisibility by 10

**A number is divisible by 10 if its ones place is only zero.** Observe that numbers divisible by 10 are also divisible by 5.

**Examples:**

1. 2020 is divisible by 10 (2020÷10 = 202) where as 2021 is not divisible by 10.
2. 26011950 is divisible by 10 and hence divisible by 5.

## Divisibility by 11

**A number is divisible by 11 if the difference between the sum of alternative digits of the number is either 0 or divisible by 11.**

**Examples:** Consider the number 256795. Here, the difference between the sum of alternative digits = (2+6+9 )−(5+7+5)=17−17=0.
Hence, 256795 is divisible by 11.



**Activity**

The teacher may ask all the students to check mentally for divisibility by 2, 3, 4, 5, 6, 8, 9, 10 and 11. If divisible, let them write 'yes', otherwise 'no'(the first one is done for you!).

| Number | ÷ 2 | ÷ 3 | ÷ 4 | ÷ 5 | ÷ 6 | ÷8 | ÷9 | ÷10 | ÷11 |
|---|---|---|---|---|---|---|---|---|---|
| 68 | yes | no | yes | no | no | no | no | no | no |
| 99 | | | | | | | | | |
| 300 | | | | | | | | | |
| 495 | | | | | | | | | |
| 1260 | | | | | | | | | |
| 7920 | | | | | | | | | |
| 11880 | | | | | | | | | |
| 13650 | | | | | | | | | |
| 600600 | | | | | | | | | |
| 15081947 | | | | | | | | | |

## 1.4 Prime Factorisation

Expressing a given number as a product of factors that are all prime numbers is called the ***prime factorisation*** of a number. For example, 36 can be written as product of factors as

36 =1 x 36; 36 = 2 x 18; 36 = 3 x 12; 36 = 4 x 9; 36 = 6 x 6

Here, the factors of 36 can be found easily as 1, 2, 3, 4, 6, 9, 12, 18 and 36. Note that not all the factors of 36 are prime numbers. To find the prime factors of 36, we do the prime factorisation by the following methods.

1. **Division Method** 2. **Factor Tree Method**



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